Three-color triangles mathematical overview

Adult theorem map | Draft and unpiloted

Setting

A large triangle is divided into finitely many little triangles meeting only along whole edges or at vertices. Its corners have labels R, B and Y. Every vertex on an outside side may use only that side's two corner labels; interior vertices may use any of the three labels. Count little cells, not larger triangles made from several cells.

Sperner parity theorem

Every such labeling has an odd number of little triangles with all three labels, and therefore at least one. This statement is independent of triangle sizes and of the particular subdivision. Allowing a forbidden boundary label can destroy the guarantee.

The door invariant

An R-B edge is a door. A little triangle has zero, one or two doors; it has exactly one precisely when it has R, B and Y. Across the outside boundary there are an odd number of doors: only the R-B side contributes, and its endpoint labels differ. Each interior door is counted twice. Thus the number of one-door cells is odd. In the equivalent path picture, two-door cells pass a route through; components may be paths or closed loops. A chosen outside entrance can return outside, although not all entrances can pair that way.

Across the bands

K-1 moves labels and searches for possible counts and locations, including a changed-boundary example. Grades 2-3 learns doors, local cases and pairing. Grades 4-5 connects boundary parity to all-three cells on any allowed triangulation and tests the boundary hypothesis.

Discovery before proof

Let children experiment before naming an odd-count conjecture. A complete check of a small board proves only that board; the door count or pairing argument explains every permitted labeling and subdivision. The detailed paths, witnesses and proof follow the preparation pages.

Three-color triangles facilitator guide

Library slot 16. Separate adult guide for F16-K-v3, F16-23-v3 and F16-45-v3. Each band has Problems 1-6. K-1 has 6 pages; each older packet has 7 pages because Problem 1 has a second recording board. Draft and unpiloted; the slot number does not set a meeting calendar.

The mathematical destination

Labels live on vertices. Boundary restrictions force an odd number of little triangles with all three labels. R-B edges, called doors, convert this guarantee into a finite path and pairing argument. An individual outside entrance can return outside; the odd number of entrances guarantees that at least one reaches an all-three cell.

Entry	Prerequisites and useful endpoint
K-1	Follow spoken side rules, place lettered counters and recognize a three-label set. No reading or arithmetic beyond small counts. Find a maximum or move one all-three cell.
Grades 2-3	Read R/B/Y, trace edges and pair objects as odd/even. Reach local door classification and a reason zero is impossible.
Grades 4-5	Track complete cases, paths and odd/even. Reach the general parity proof and test the importance of its boundary hypothesis. No fixed-point theory is needed.

Prepare for eleven children

Print selected pages single-sided on US Letter at 100%. Provide eleven sets of three colored pencils, graphite pencils/erasers and at least 20 spare sheets. For the four K-1 children, prepare four counter sets, each with 15 R, 15 B and 15 Y counters: 180 total. Counters about 3/4 inch across fit every K-1 board, whose tightest spacing is 1.3125 inches, and cover its 0.3-inch rings; red, blue and yellow counters with a marker letter on each work. These are preparation targets, not verified inventory. Put letters or distinct symbols on every counter; do not rely on color. The largest youngest board has 15 vertices, so test counter clearance on a printout.

Keep three stable adult-led tables and choose by readiness. Keep corners fixed; enforce the permitted pair on the entire outside side. Every blank side dot is split in the tints of the two letters its side allows (pink and blue along the bottom, pink and yellow on the left, blue and yellow on the right), and in three tints where a rule is relaxed: the K-1 Problem 6 bottom and the Grades 4-5 Problem 6 star. The printed words beside each side state the same rule; letters, not tints, decide. Leave cell interiors and edge midpoints clear for tracing. A little triangle is a cell with no subdivision inside it; larger triangles spanning several cells do not count. The fan board has four extra vertices inside the four original cells, and 12 little cells.

Before teaching, label a board, count only its little cells and trace an R-B door route. Give older children the two Problem 1 copies to preserve a minimum and maximum. Use pencil changes or movable counters before asking for a general argument.

Flexible one-hour menu

Minutes	Use
0-5	Running around on arrival.
5-9	Handle counters or color a practice triangle before introducing restrictions, then the shared launch below.
9-25	K-1 P1-P3; middle P1-P2; upper P1-P2. Let the search last.
25-47	Choose a route: K-1 P4 or P5/P6; middle P3-P4 then P5 or P6; upper P3-P5 or the boundary exception P6. Once a table has done its rows problem (P4), a one-minute stand-up game: switch between R and B gestures and notice when the end matches the start.
47-52	Share one surprising filling and one reason. No band must finish every page.

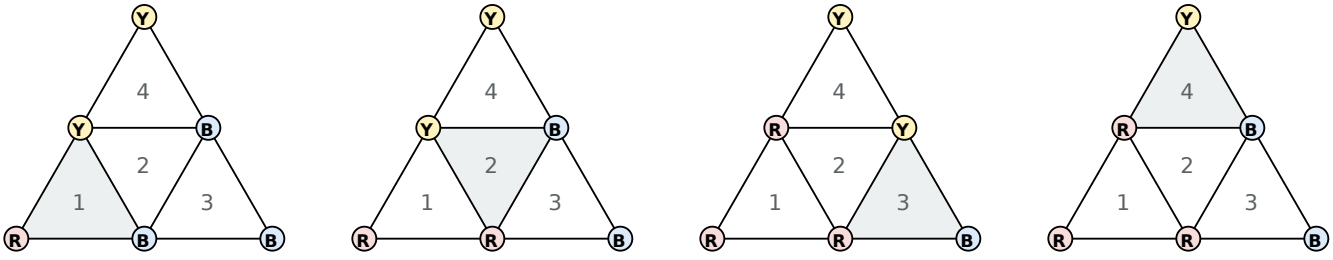
Shared launch

On the K-1 Problem 1 page (the two-step board), or a whiteboard copy of it, place the three fixed corner counters and let children fill the remaining vertices legally. Point to a little triangle and check its three vertex labels. Change one counter without coloring the face. Ask, “Can you move an all-three triangle somewhere else?” Demonstrate the side restrictions with one illegal counter and have children repair it, pointing to the two tints of that side dot. Save the theorem and door technique for after experimentation.

K-1 Problem 1

Each of the four positions can contain the unique all-three cell, one at a time. The four fillings below are exact witnesses. All eight legal two-step fillings have exactly one; the request is sequential, not to make all four work simultaneously. Hint: keep the target triangle’s needed labels and repair the remaining boundary vertices.

Cell 1 **Cell 2** **Cell 3** **Cell 4**



In guide diagrams, small gray numbers name cells; light shading marks the all-three cells. Letters, not shading, determine correctness. Numbering runs left-to-right within each horizontal strip, then upward (the fan board is numbered region by region: lower-left, central, lower-right, top). It is adult notation, not an added student requirement.

Hold hints back

First ask a child to show what they tried. Offer a hint only after another attempt. Ask about one edge or one small region rather than supply a full filling. Give an oral or physical explanation full credit when it accounts for all cases.

K-1 Problems 2 to 4

Problem 2

No legal filling has zero all-three cells. The side rules are the hypotheses of the parity proof on page 11. For K-1, preserve time to try and notice that changing one cell can create another. If they want a reason, let the adult mark R-B doors and pair outside entrances: an odd number cannot all be paired with outside exits. Do not present “we tried many” as a proof.

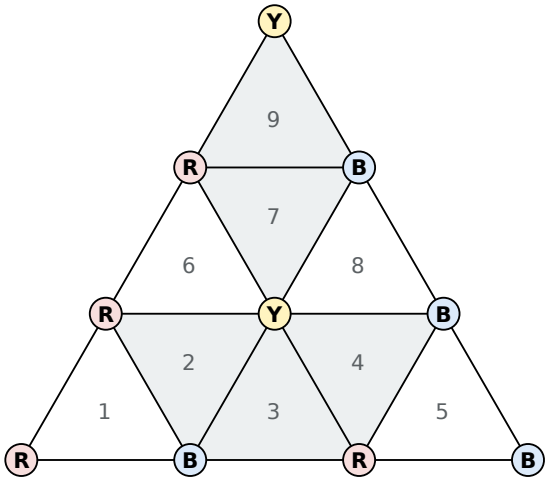
Problem 3

The maximum is 5 on the nine-cell, three-step board. A witness appears below: bottom-to-top row code RBRB/RYB/RB/Y. Its all-three cells are 2,3,4,7,9. Page 5 proves no legal filling has more than 5. Hint: once a child finds three, ask whether changing the center could help and what that change might destroy.

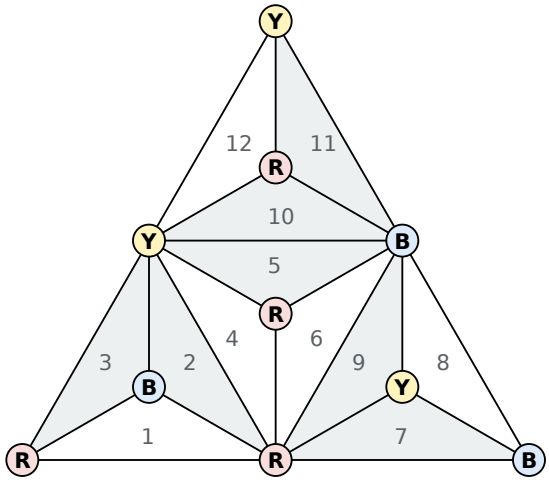
Problem 4

Maximum 7. Preserve the six printed labels: bottom RRB, next row YB, top Y. Fill the four inserted centers, by location, lower-left B; central inverted region any of R/B/Y; lower-right Y; top R. There are exactly three maximum fillings, differing only at the central inserted dot. The drawing uses R there.

Problem 3 maximum 5



Problem 4 maximum 7

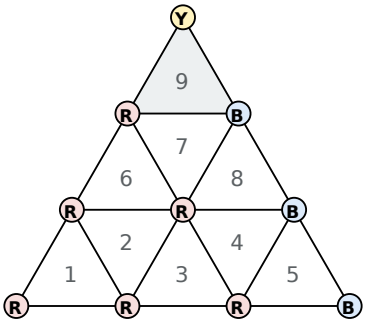


Proof: the original lower-left triangle has corner labels RRY; its three subcells can contribute at most 2, attained by a B center. The original lower-right RBB region contributes at most 2 with Y. The top YBY region contributes at most 2 with R. The central RBY region contributes exactly 1 for any center label: only the edge joining the other two labels makes a three-label subcell. Thus the bound is 2+2+2+1=7, and the displayed filling attains it. Hint: inspect the four original regions separately, while leaving their printed vertices fixed.

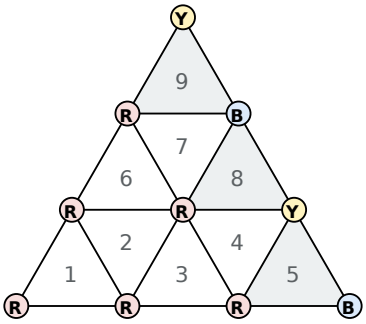
The three-step minimum and maximum

Grades 2-3 Problem 1 and grades 4-5 Problem 1: the minimum is 1, the maximum 5, and the only attainable counts are 1,3,5. The three exact witnesses below use row codes RRRB/RRB/RB/Y; RRRB/RRY/RB/Y; RBRB/RYB/RB/Y, respectively. The second student page is another board for Problem 1, not a new problem.

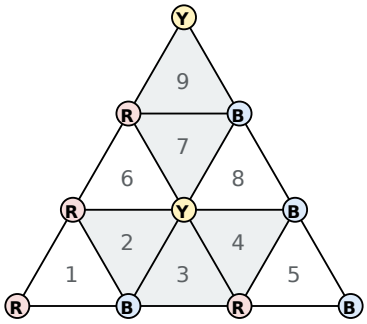
1 all-three cell



3 all-three cells



5 all-three cells



Why five is the upper bound

Here is a finite structural proof, independent of the enumeration. Rotate the board and rename the three colors together so that the interior center is Y and the opposite bottom corners are R and B. This preserves the problem. Write L,M for the two noncorner bottom labels, left to right. They are each R or B. Write U for the boundary label just below the top corner on the right, and V for the one on the left. U is B or Y; V is R or Y.

Cells 6 and 8, along the left R/Y and right B/Y edges, both lack one required color. The bottom inner cell contributes 1 exactly when L differs from M. The two cells meeting at the lower-left corner region together contribute 1 exactly when L=B; the lower-right pair together contribute 1 exactly when M=R. Each of the two cells along the top inner edge contributes 1 exactly when U=B and V=R.

These groups partition all nine cells: two always-zero cells, one bottom inner cell, two lower-left cells, two lower-right cells, and two top-edge cells. So the total equals a bottom contribution from the table, plus either 0 or 2 from the top-edge pair.

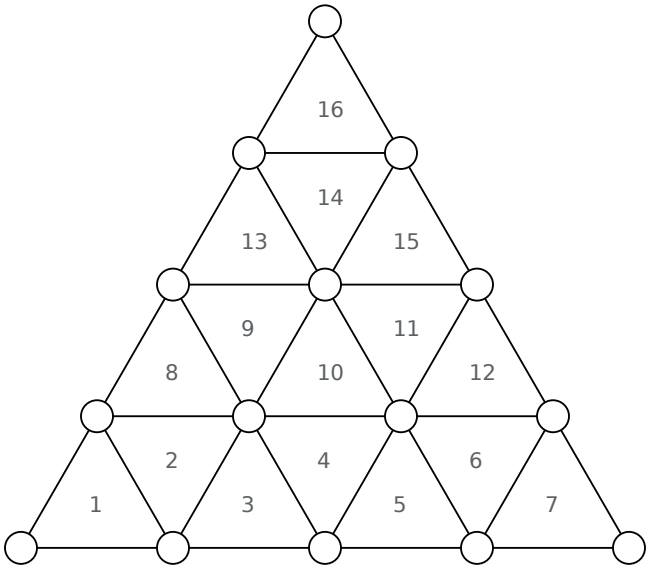
L	M	Bottom contribution
R	R	1
R	B	1
B	R	3
B	B	1

The total is therefore 1,3 or 5. The witnesses attain all three. This proves both the upper and lower bounds without appealing to the general theorem. Independent enumeration checks the distribution: 108 legal fillings have 1, 72 have 3, and 12 have 5, among 192 legal fillings.

Held hint: Preserve the best two records, then track what one interior change gains and loses. Ask the upper band whether an even total survives.

K-1 Problem 5 all sixteen locations

Every one of the 16 small cells can be the only all-three cell. Each table entry is a checked witness for the numbered board below. Read each slash-separated row left-to-right, starting with the bottom row of five vertices, then four, three, two and the top Y. For example RRRRB/RRRB/RRB/RB/Y makes only cell 16 all-three.



Cell	Bottom-to-top row code	Cell	Bottom-to-top row code
1	RBRRB/YBRB/YBB/YB/Y	9	RRRRB/RRRB/YBB/YB/Y
2	RRRRB/YBRB/YBB/YB/Y	10	RRRRB/RRBB/RYB/RY/Y
3	RRBRB/RYBB/RYB/RY/Y	11	RRRRB/RRRB/RYB/RY/Y
4	RRRRB/RYBB/RYB/RY/Y	12	RRRRB/RRRB/RRY/RY/Y
5	RRRBB/RRYB/RRY/RY/Y	13	RRRRB/RRRB/RBB/YB/Y
6	RRRRB/RRYB/RRY/RY/Y	14	RRRRB/RRRB/RRB/YB/Y
7	RRRRB/RRRY/RRY/RY/Y	15	RRRRB/RRRB/RRB/RY/Y
8	RRRRB/RBRB/YBB/YB/Y	16	RRRRB/RRRB/RRB/RB/Y

The table proves all sixteen locations are possible by explicit construction; there are no other locations because the board has exactly sixteen cells. Do not expect a child to finish sixteen fillings in the hour. A child can keep a record of which cells have worked and return later. Hint: move a boundary between label regions a little at a time, then check that it did not create a second all-three cell. Avoid moving fixed corner letters.

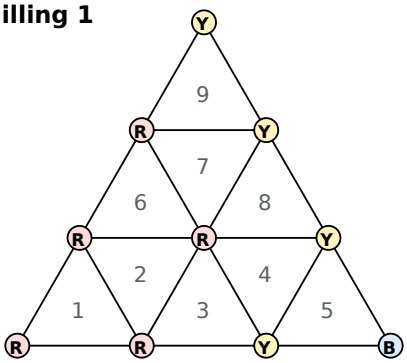
Independent enumeration found 2,920 legal fillings with one all-three cell. The whole four-step distribution is 1,3,5,7,9 all-three cells in 2,920; 6,192; 3,840; 848; 24 fillings, respectively. These checks support this board; they are not a proof for arbitrary triangulations.

When the boundary rule changes

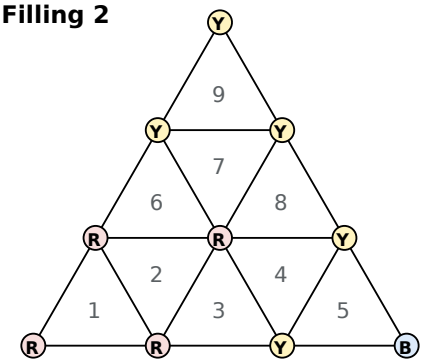
K-1 Problem 6

Four different zero-cell fillings are shown below. Their row codes, in the same order, are RRYB/RRY/RY/Y; RRYB/RRY/YY/Y; RRYB/RYB/RY/Y; RRYB/RYB/YB/Y. Every corner remains fixed. Only the noncorner bottom labels gained permission to use Y. In all four examples the bottom is R-R-Y-B, so it has no R-B door. Hint: try using the newly allowed Y to separate the bottom R and B regions. An adult letters each find on a small triangle below the board.

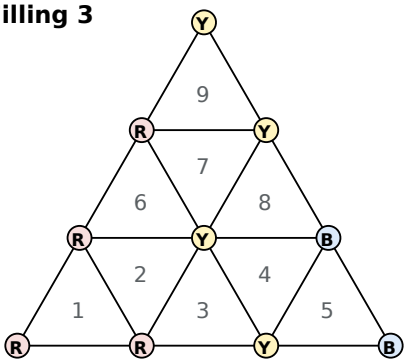
Filling 1



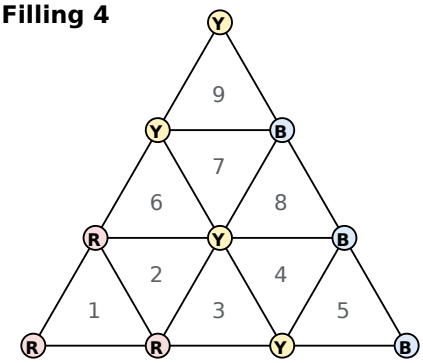
Filling 2



Filling 3

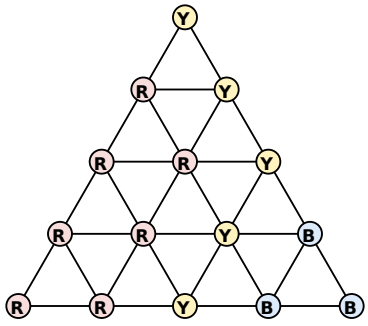


Filling 4



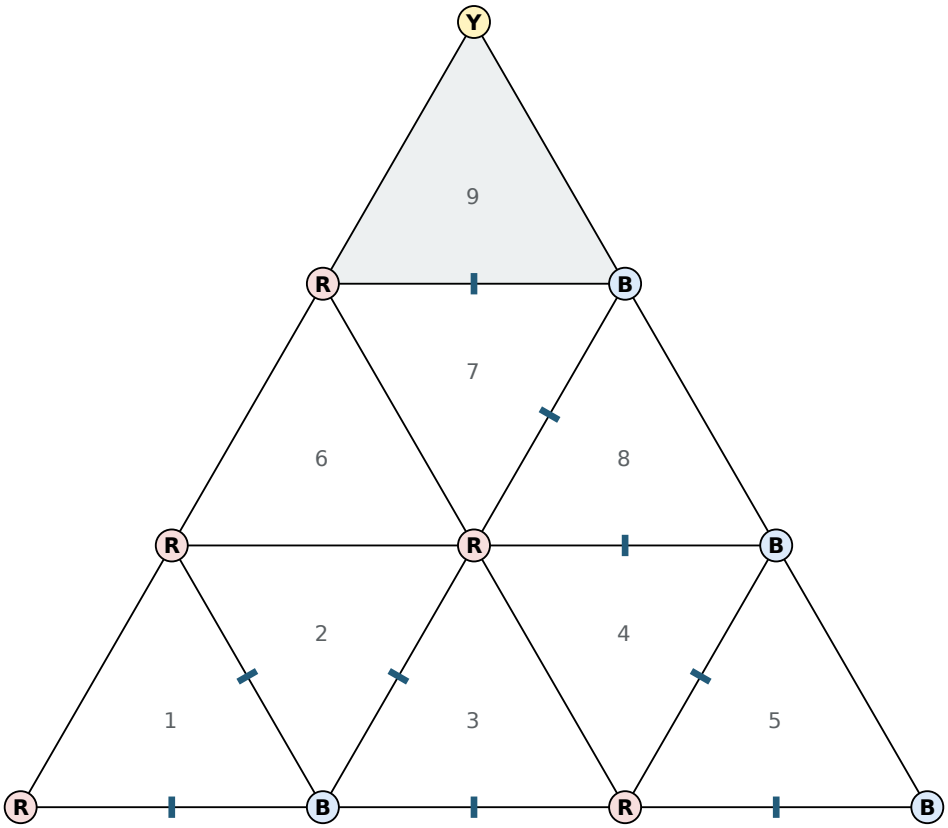
Grades 4-5 Problem 6

A zero-cell witness is RRYBB/RRYB/RRY/RY/Y, shown below. The third bottom label is the starred midpoint Y; every other side rule remains legal. This does not contradict Problem 5 because its R/B-only bottom-side hypothesis is false. The bottom sequence R-R-Y-B-B has no R-B door, so the odd-boundary-door proof no longer applies. There are 208 zero-cell fillings with this one-dot exception, but one verified witness answers the task. Hint: locate the exact sentence in the earlier theorem that the star changes.



Grades 2-3 Problem 2 doors and routes

The printed filling is RBRB/RRB/RB/Y. It has 9 doors, including the three bottom edges. In the solution diagram, a short dark blue bar crosses each door. Cell 9 is the only all-three cell. Call the three bottom segments entrances 1,2,3 from left to right.



All boundary-started routes, up to reversing the same route, are:
Entrance 1 - cells 1,2,3 - entrance 2.
Entrance 3 - cells 5,4,8,7,9, stopping in cell 9.

Thus entrances 1 and 2 lead back outside; entrance 3 reaches the all-three cell. Starting separately from entrances 1 and 2 traverses the same component in opposite directions. This is why choosing any one entrance need not find an all-three cell. There are no additional internal components on this printed board.

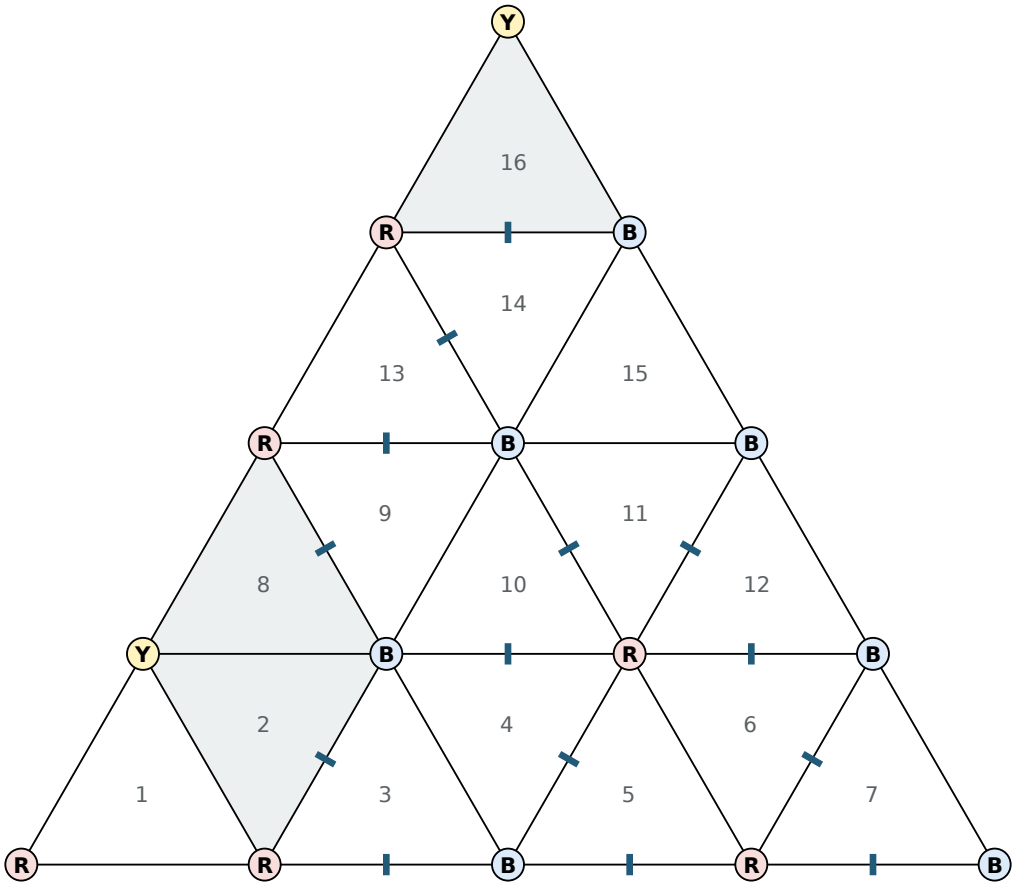
Grades 2-3 Problem 5

For exactly three all-three cells on the four-step board, use the upper printed filling on page 9 of this guide: RRBRB/YBRB/RBB/RB/Y. Cells 2,8,16 are all-three. Its three bottom doors are segments 2,3,4. The boundary routes are segment 2 - cells 3,2, and segment 3 - cells 5,4,10,11,12,6,7 - segment 4. The latter two entrances pair up, but segment 2 ends inside.

No legal filling can have every outside route return outside. Outside-to-outside components consume entrances in pairs, while the number of boundary doors is odd. The internal route between cells 8 and 16 need not be traced for this middle-band problem, which asks only for outside-started routes. Hint: count entrances before tracing, then see which are paired.

Grades 4-5 Problem 2 complete door components

The printed row code is RRBRB/YBRB/RBB/RB/Y. All 14 doors are marked. There are three boundary doors, on bottom segments 2,3,4, and three all-three cells, numbered 2,8,16.



There are exactly three route components, up to reversal:
1. Bottom segment 2 - cells 3,2. It ends at all-three cell 2.
2. Bottom segment 3 - cells 5,4,10,11,12,6,7 - bottom segment 4. It returns outside.
3. Cells 8,9,13,14,16. It stays inside and joins all-three cells 8 and 16.

The internal component is an open path, not a loop. None of the three components branches. Cell 1 has no door and is not part of a route. A component traced from its other end is the same route in reverse. These lists account for all 14 doors: respectively 2,8 and 4 doors.

The answer to “Can every route from outside return outside?” is no, both on this board and under the legal boundary rules generally. The three entrances here cannot all pair with outside exits. More generally, the proof is on page 11. Hint: after tracing outside components, search for an unused door entirely inside the board.

Local fillings and boundary rows

Problem 3 in both older bands

There are ten classes, with turns and flips identified but color names retained. Repeated letters are allowed. Three monochromatic cases, six two-color cases and one three-color case exhaust all multisets of three labels. The twelve printed triangles include two spare spaces.

Door count	Complete fillings
0	RRR, BBB, YYY, RRY, RYY, BBY, BYY
1	RBY
2	RRB, RBB

Only RBY has exactly one door. If both R and B occur and Y does not, one label appears twice; each copy connects to the other label, giving two doors. If either R or B is absent, there are none. Three doors are impossible: after putting R and B at two vertices, the third label is R (two doors), B (two doors) or Y (one door). Hint: sort by how many different letters are used, then decide which repeated-letter choice is new.

Grades 2-3 Problem 4

All five rows must have an odd number of doors, regardless of the empty labels. The rows have 2,3,4,5,6 edges, so the possible counts are respectively {1}, {1,3}, {1,3}, {1,3,5}, {1,3,5}. For a one-door filling, use R until the final B. More generally, any chosen set of k change positions with k odd determines a legal row from R to B, so every listed count is attainable.

Grades 4-5 Problem 4

Top to bottom, the rows can have an odd, even, odd, even and odd number of doors, and every count of that parity up to the number of edges occurs. The rows have 3,4,5,6,7 edges; possible counts are {1,3}; {0,2,4}; {1,3,5}; {0,2,4,6}; {1,3,5,7}. Equal neighboring labels are not doors. Each door switches the current label. An even number of switches returns to the starting label; an odd number ends at the other label. This applies to any length, including a row with no changes.

Sample row fillings, top to bottom, are RRRB; RRRRR; RRRRRB; RRRRRRR; RRRRRRRB. They show the required parity with the fewest doors. To make more, insert a B region inside an R region, adding two changes while leaving the endpoints fixed. Hint: have the child start with an R counter and flip it at each marked door.

The local-to-global bridge

Triangles with two doors pass a path on; triangles with one door stop it; triangles with no doors do not join the door network. This classifies what a route can do before any general proof. A finite route entered from outside cannot get trapped in a disjoint cycle, because every interior transit cell has exactly two doors and provides no branch into a cycle.

Why an all-three cell cannot disappear

Grades 2-3 Problem 6 and grades 4-5 Problem 5

No zero-cell filling is possible on the fan board in middle P6. Upper P5 asks for more: on every finite edge-to-edge triangulation of the large triangle with the stated boundary labels, the number of all-three cells is odd. The cell shapes and sizes need not match. There must be no holes, gaps or unmatched T-junctions.

Along the bottom side, the labels begin R, end B and use only those two letters. The row argument gives an odd number b of boundary doors. Neither other side has a door: the left lacks B and the right lacks R. Every all-three cell has one door; every other cell has zero or two.

Count door incidences, once from each adjoining cell. If T counts all-three cells, U counts two-door cells and I counts interior doors, then $T+2U=b+2I$. The left side counts by cells. On the right, an interior door is counted twice and a boundary door once. Thus T has the same parity as b, which is odd. In particular T is at least one. This supplies a proof for every filling, not merely the checked examples.

Without symbols, pair the two marks belonging to each interior door and the two doors belonging to each two-door triangle. What remains on one side are the all-three triangles and on the other side the boundary doors. Adding or removing pairs never changes odd/even, so both leftover counts are odd.

The path version and its important qualification

The finite door network consists of nonbranching paths and possibly closed cycles. Its endpoints are boundary entrances and all-three cells. Paths joining two boundary entrances pair them; paths joining two all-three cells pair those. Each remaining path joins one of each. Since the boundary endpoints are odd, there are an odd number of these mixed paths, and hence an odd total of all-three endpoints. A chosen entrance may return outside, so “start anywhere and you must find one” is not valid.

Held hints: First revisit the ten local types. Then count the boundary changes and ask which endpoints are paired by complete routes. Only after that offer the incidence equation. For K-1, let paired entrances be physical counters; the adult can carry the full argument.

Extensions and fidelity limits

Change a side label and find which part of the proof fails; use the exact exceptions on page 7. Subdivide one little cell by adding an interior vertex and compare old/new parity. This is Sperner’s triangulation lemma, distinct from the antichain theorem with the same name. No claim about actual rent allocation, limits or fixed points follows from this packet alone.

Sources and verification

Francis E. Su, “Rental harmony: Sperner’s lemma in fair division,” American Mathematical Monthly 106 (1999), 930-942, author PDF pp. 2-5, especially the door-path and incidence arguments: https://math.hmc.edu/su/wp-content/uploads/sites/10/2019/06/Rental-harmony.pdf. Explicit two-dimensional statement: Harvey Mudd Math Fun Facts, https://math.hmc.edu/funfacts/sperners-lemma/. The diagrams, exact instances and row-code keys here were independently computed for the final packet.

Teaching evidence: Rozhkovskaya, Math Circles for Elementary School Students, “Introduction: Berkeley 2009,” describes individual adult support. This teaching menu is unpiloted. Independent checking rebuilt cells by geometric adjacency, checked area and shared-edge incidences, and verified distributions, witnesses and door routes. Editable checks accompany this guide. Two door examples were added.

Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

Use and return choices

On another visit, K-1 can move a single all-three cell (P5), optimize the subdivided board (P4), or change the boundary rule (P6). Grades 2-3 can return to the local door cases and two-color rows (P3-4) before the impossibility argument (P5-6). Grades 4-5 can keep their minimum/maximum fillings and build the parity explanation (P3-5), with P6 as a separate test of the boundary hypothesis. A first visit ending with a well-checked filling leaves substantial mathematics for the next.

Current youngest table

For four K-1 children, prepare four sets of fifteen counters in each of R, B and Y: 180 counters altogether. Two shared pair sets need 90. Keep the prescribed counter size and print scale; actual stock fit remains untested.